BRAC IT CodeSprint 2026

Technical Specification: BRAC স্বাস্থ্যকর্মী প্রশিক্ষণ – AI সিমুলেটর

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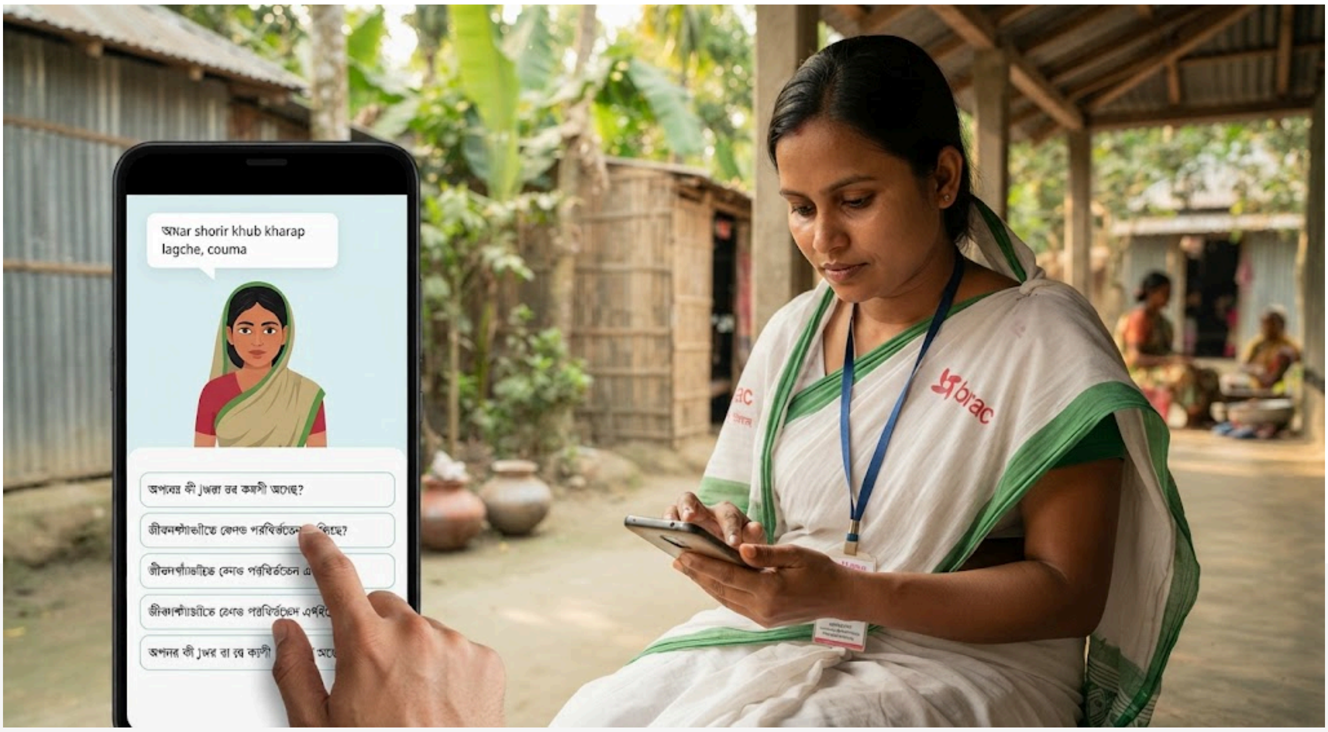
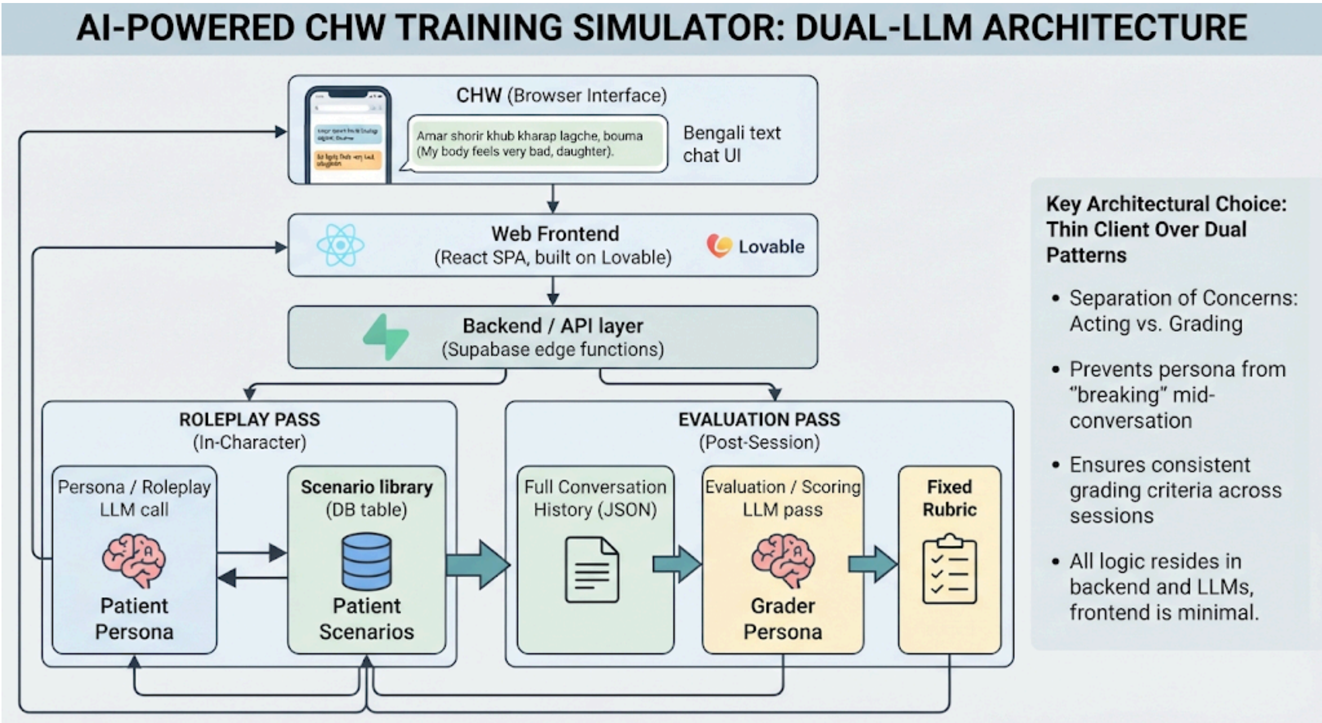


Live Demo: https://health-mentor-sim.lovable.app/

1. System Purpose

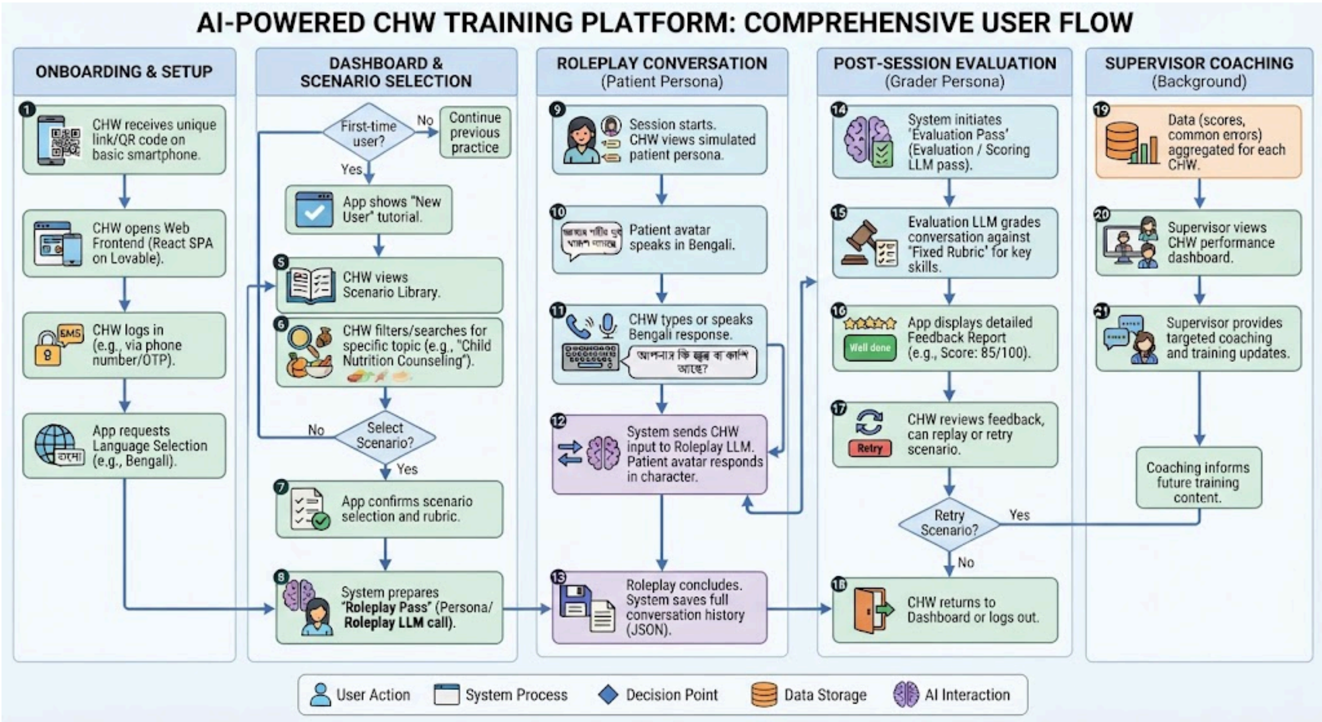
BRAC Health Mentor Sim is an AI-powered conversational training tool for BRAC's community health workers (CHWs). It replaces static training scripts with a live, language-model-driven patient simulation: the CHW conducts a real consultation in Bengali against an AI "patient," and receives structured, rubric-based feedback afterward. The goal is to let workers rehearse difficult or rare case types (non-compliant patients, culturally sensitive topics, emergency triage) as many times as needed, at zero marginal cost per session, before facing them in the field.

2. High-Level Architecture



The system is a thin client over two distinct LLM invocation patterns rather than one continuous chat: a **roleplay pass** that stays in character as the patient, and a **post-session evaluation pass** that steps outside the roleplay to grade the CHW's performance against a fixed rubric. Keeping these as separate calls (rather than asking one prompt to both act and grade) avoids the persona "breaking" mid-conversation to give feedback, and keeps grading criteria consistent across sessions.

4. User Flow



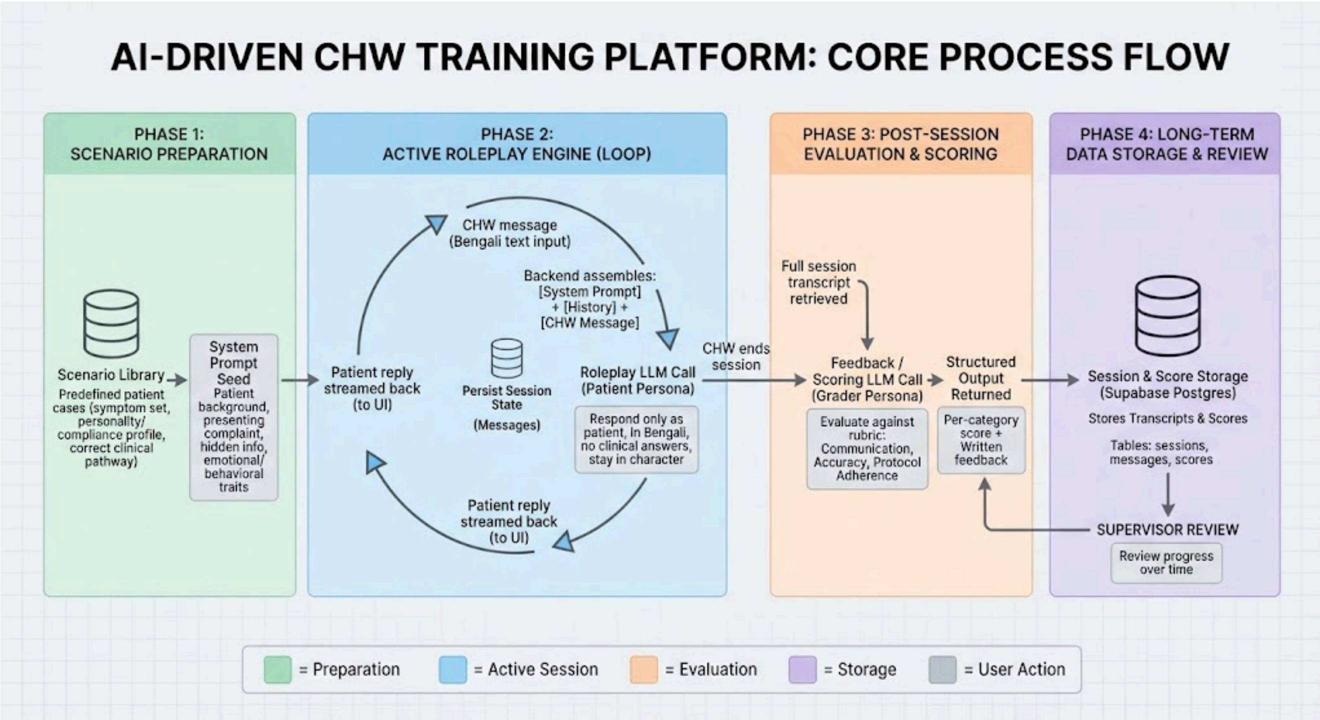
3. Tech Stack

Layer	Choice	Rationale
Frontend	React + TypeScript SPA (Lovable-generated scaffold)	Fast iteration for a CodeSprint timeline; no native app needed
Backend	Supabase (Postgres + Auth + Edge Functions)	Pairs natively with Lovable; handles auth, storage, and serverless LLM calls without standing up separate infra
LLM	Hosted model via API (model-agnostic at the interface layer)	Keeps the persona/feedback prompts portable if the underlying model is swapped for cost or quality reasons
Language handling	Bengali prompts and system instructions, with the model responding natively in Bengali	Avoids a translation layer, which would add latency and lose idiomatic nuance in patient dialogue
Hosting	Lovable-managed deployment (health-mentor-sim.lovable.app)	Zero-ops for a hackathon-stage prototype

4. Core Components

4.1 Scenario Library A set of predefined patient cases (symptom set, personality/compliance profile, correct clinical pathway) stored as structured records. Each scenario seeds the roleplay system prompt with: patient background, presenting complaint, hidden information the CHW must elicit through good questioning, and emotional/behavioral traits (anxious, dismissive, non-compliant, etc.).

4.2 Roleplay Engine On each CHW message, the backend assembles: the scenario's system prompt + conversation history + the CHW's latest message, and sends it to the LLM with instructions to respond only as the patient, in Bengali, without breaking character or revealing clinical answers directly. Session state (message history) is persisted per session so the model has full context on each turn.

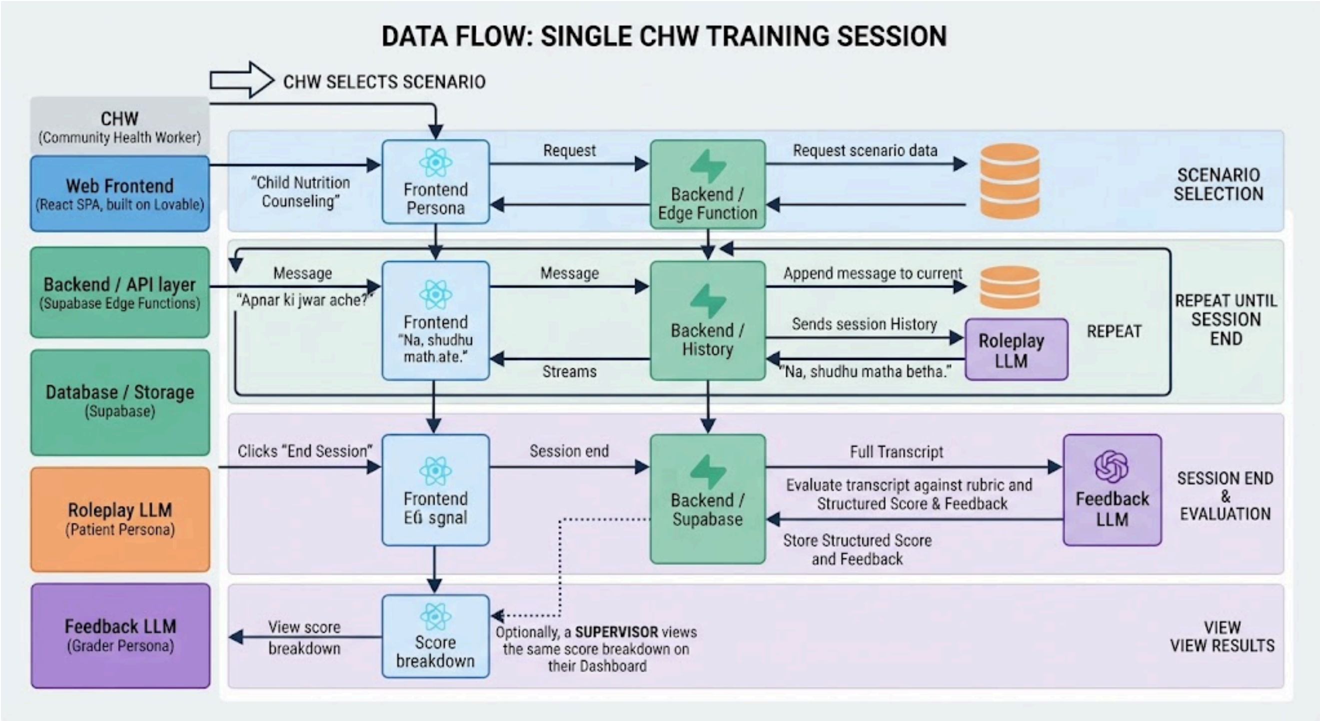


4.3 Feedback / Scoring Engine Once the CHW ends the session, the full transcript is passed to a second LLM call with a different system prompt: an evaluation rubric covering communication (empathy, clarity), clinical accuracy (correct questions asked, correct advice given), and protocol adherence (referral/escalation triggers). The model returns structured output (e.g., per-category score + short written feedback) rather than free text, so it can be rendered consistently in the UI and, if needed, aggregated across a worker's session history.

4.4 Session & Score Storage Transcripts and scores are persisted (Supabase Postgres tables: `sessions`, `messages`, `scores`) so a worker's supervisor can review progress over time rather than only seeing a single session's result.

5. Data Flow (Single Session)

1. CHW selects a scenario → frontend requests scenario data.
2. CHW sends a message → edge function appends to session history → roleplay LLM call → patient reply streamed back.
3. Repeat until CHW ends the session.
4. Full transcript → feedback LLM call → structured score returned and stored.
5. CHW (and optionally supervisor) views the score breakdown.



6. Key Technical Considerations

- **Persona consistency:** the system prompt fixes the patient's personality and undisclosed information explicitly, so the model doesn't invent contradictory details across turns or accidentally reveal the "correct answer" to the CHW.
- **Bengali-first design:** prompts, scenario content, and expected outputs are all authored in Bengali rather than translated post-hoc, since CHWs will practice in the language they use in the field.
- **Separation of roleplay and grading:** using two distinct prompts/calls (rather than one) prevents the grading logic from leaking into the in-character conversation.
- **Cost/latency:** each turn is a single LLM call with bounded context (scenario + recent history), keeping response times low enough for a natural back-and-forth conversation.
- **Data sensitivity:** although scenarios are synthetic, transcripts resemble clinical dialogue — stored in Supabase with row-level access control so only the CHW and authorized supervisors can view a given session.

7. Known Limitations / Future Work

- Scenario library is currently static; a future version could let supervisors author new cases directly.
- Feedback is currently per-session; longitudinal analytics across a worker's full training history is a natural next step.
- Voice input/output (rather than text-only chat) would more closely mirror real consultations and is a candidate for a later phase.
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